

AI Lab Test-03

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Q1: RETAIL SECTOR – WEB FRONTEND DEVELOPMENT

PROMPT:

Create a simple responsive product listing webpage using HTML, CSS, and JavaScript with search, filter, and cart features.

SOURCE CODE:

```
<!DOCTYPE html>
<html>
<head>
<title>Retail Product List</title>
<style>
body{font-family:Arial;background:#f4f4f9;margin:20px}
.card{background:#fff;padding:10px;border-radius:8px;box-shadow:0 2px 5px
#ccc;text-align:center}
.products{display:grid;grid-template-columns:repeat(auto-
fit,minmax(200px,1fr));gap:15px}
.filters{margin-bottom:15px}
</style>
</head>
<body>
<h2>📦 Product Listing</h2>
<div class="filters">
<input id="search" placeholder="Search...">
<select
id="category"><option>All</option><option>Clothing</option><option>Electro
nics</option></select>
</div>
<div class="products" id="list"></div>
<h3>🛒 Cart</h3><ul id="cart"></ul>

<script>
const items=[{name:'Shirt',price:1200,cat:'Clothing'},
{name:'Laptop',price:50000,cat:'Electronics'},
{name:'Headphones',price:1500,cat:'Electronics'}];
let cart=[];
function show(p){
```

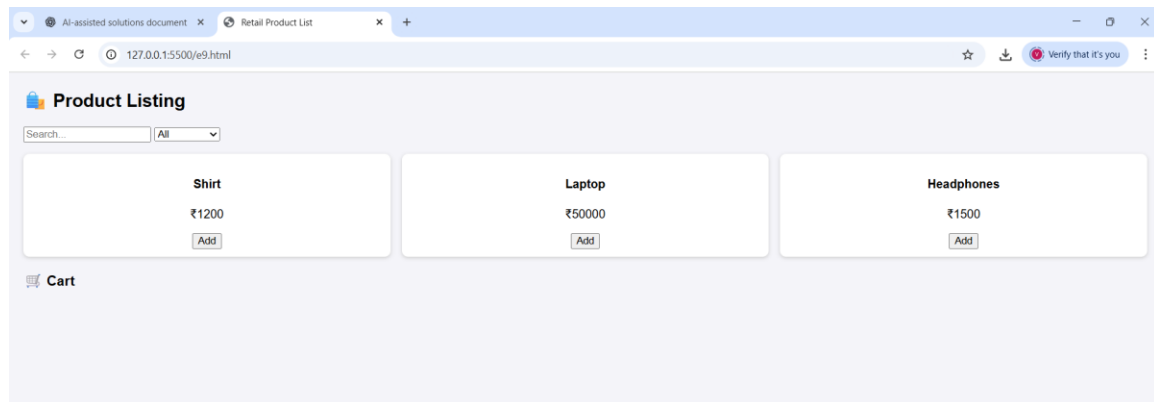
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```
document.getElementById('list').innerHTML=p.map(x=><div class='card'>
<h4>${x.name}</h4><p>₹${x.price}</p>
<button
onclick='add("${x.name}",${x.price})'>Add</button></div>`).join('');
}
function
add(n,p){cart.push({n,p});document.getElementById('cart').innerHTML=cart.m
ap(c=><li>${c.n}-₹${c.p}</li>`).join('');}
function filter(){
let s=search.value.toLowerCase(),c=category.value;
show(items.filter(i=>(i.name.toLowerCase().includes(s))&&(c==='All' || i.cat
===c)));
}
search.oninput=filter;category.onChange=filter;show(items);
</script>
</body>
</html>
```

OBSERVATION:

AI-assisted tools like ChatGPT were used to generate the complete responsive HTML, CSS, and JavaScript code. The code allows product search, category filtering, and cart management, improving development efficiency and usability.

OUTPUT:



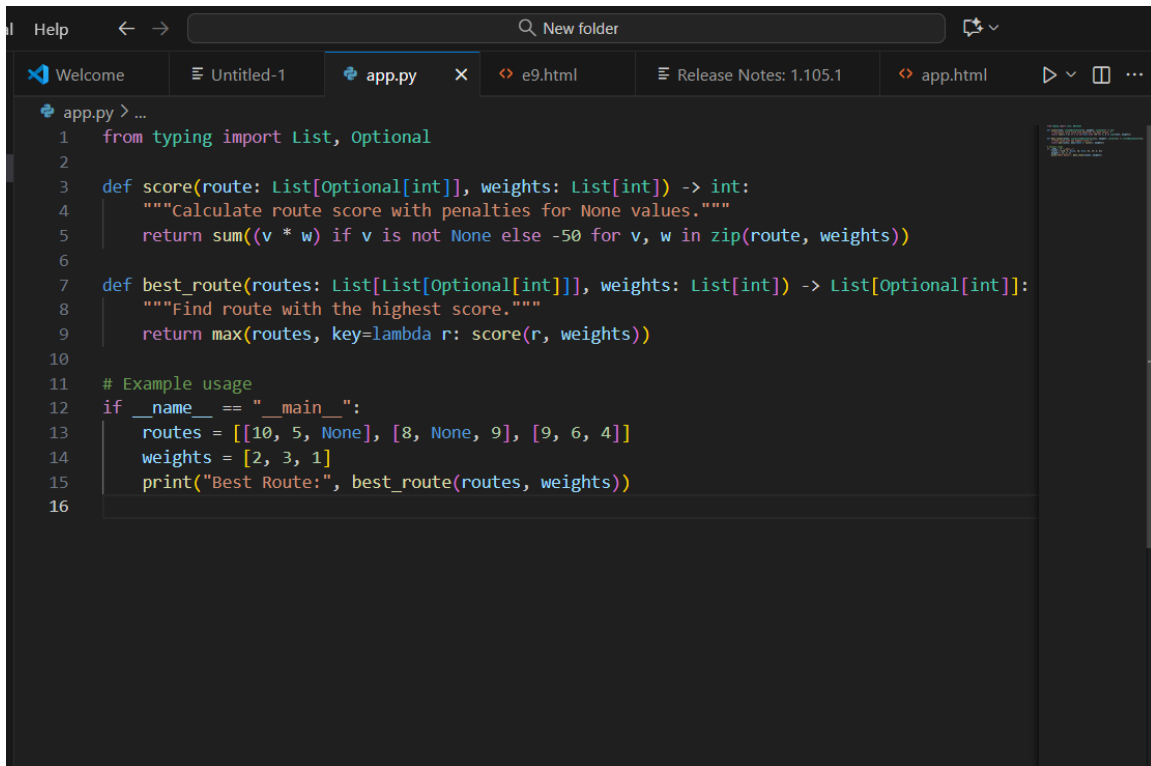
Q2: TRANSPORTATION SECTOR – CODE REFACTORING

PROMPT:

Refactor a Python route scoring function with comments, type hints, and modular structure for better readability and maintainability.

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SOURCE CODE:

A screenshot of a code editor window with a dark theme. The editor shows a Python file named 'app.py' with the following code:

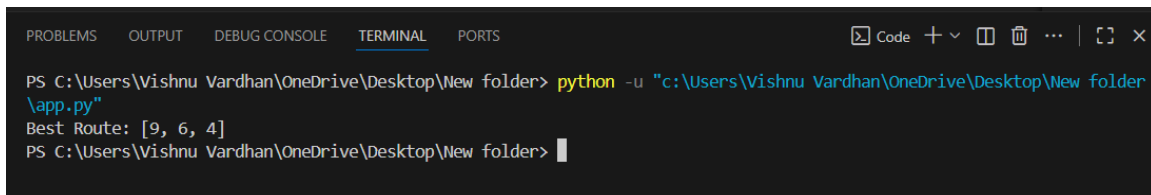
```
1 from typing import List, Optional
2
3 def score(route: List[Optional[int]], weights: List[int]) -> int:
4     """Calculate route score with penalties for None values."""
5     return sum((v * w) if v is not None else -50 for v, w in zip(route, weights))
6
7 def best_route(routes: List[List[Optional[int]]], weights: List[int]) -> List[Optional[int]]:
8     """Find route with the highest score."""
9     return max(routes, key=lambda r: score(r, weights))
10
11 # Example usage
12 if __name__ == "__main__":
13     routes = [[10, 5, None], [8, None, 9], [9, 6, 4]]
14     weights = [2, 3, 1]
15     print("Best Route:", best_route(routes, weights))
16
```

The editor interface includes a top bar with 'Help', navigation arrows, a search bar, and a 'New folder' button. Below the top bar are tabs for 'Welcome', 'Untitled-1', 'app.py', 'e9.html', 'Release Notes: 1.105.1', and 'app.html'. The 'app.py' tab is active, showing the code. A right sidebar is partially visible, showing a file explorer.

OBSERVATION:

AI-assisted refactoring improved code readability, added modularity with helper functions, and introduced type hints. The structure became cleaner, easier to debug, and more maintainable while preserving functionality.

OUTPUT:

A screenshot of a terminal window with a dark theme. The terminal shows the command to run the Python script and its output:

```
PS C:\Users\Vishnu Vardhan\OneDrive\Desktop\New folder> python -u "c:\Users\Vishnu Vardhan\OneDrive\Desktop\New folder\app.py"
Best Route: [9, 6, 4]
PS C:\Users\Vishnu Vardhan\OneDrive\Desktop\New folder>
```

The terminal window has a top bar with tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. The 'TERMINAL' tab is active, showing the command and output. The output is 'Best Route: [9, 6, 4]'. The terminal window also includes a right sidebar with icons for 'Code', '+', '-', '[]', '[]', '...', '[]', and 'X'.